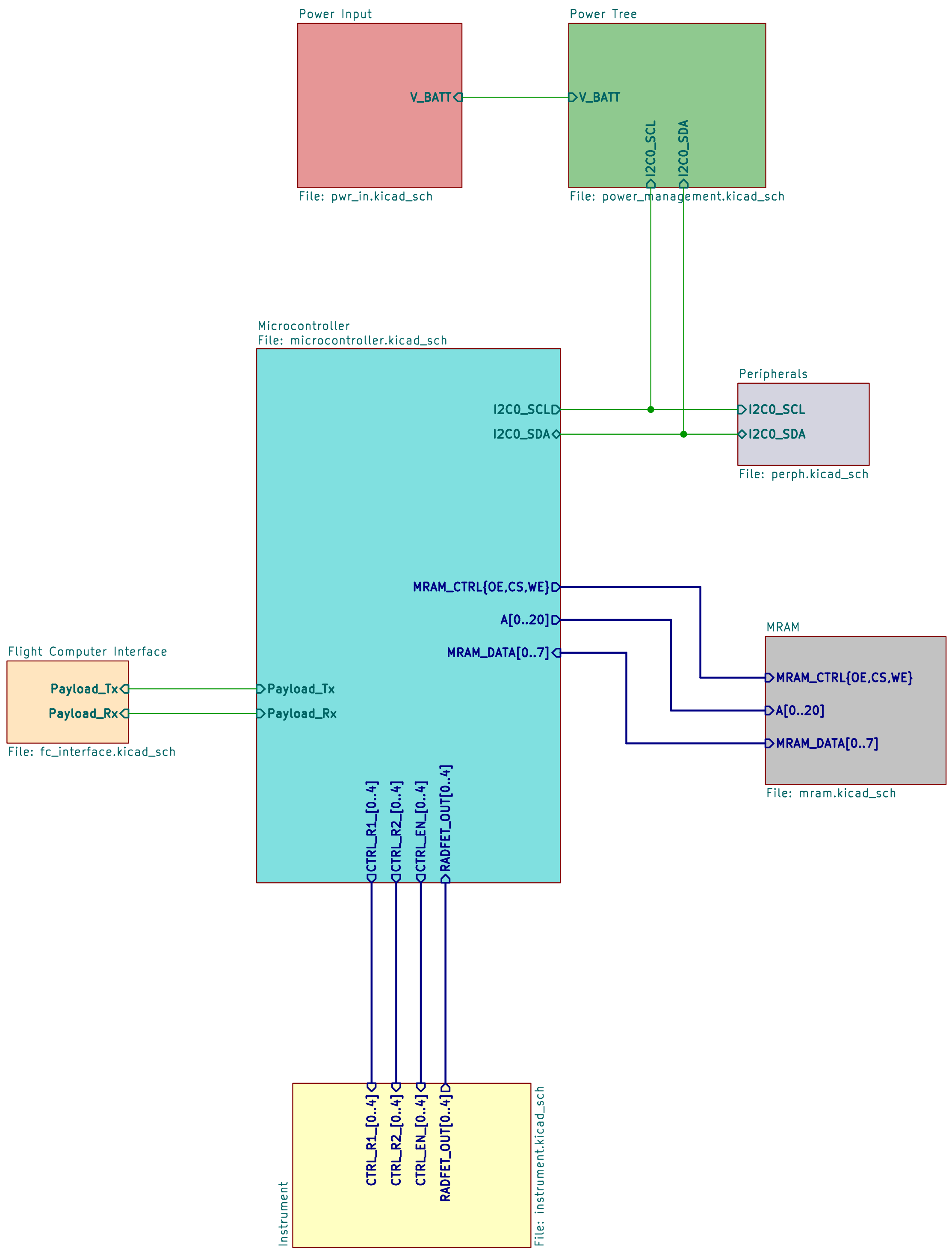

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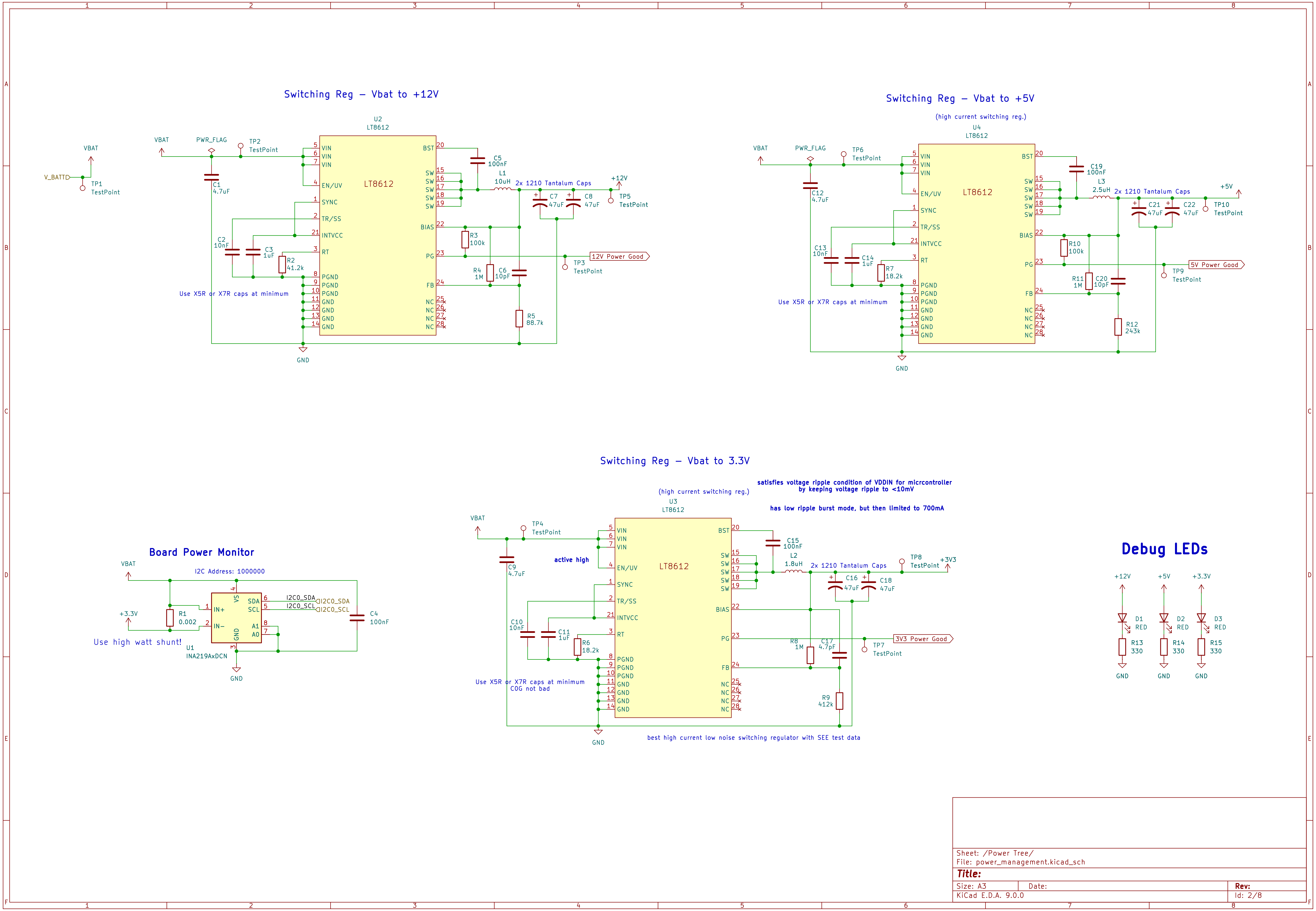
File: Payload_Controller_EM.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.0 | Id: 1/8


Switching Reg - Vbat to +12V
U2
LT8612
VBAT
PWR_FLAG
TP2
TestPoint
V_BATT
TP1
TestPoint
C1
4.7uF
C2
10nF
C3
1uF
R2
41.2k
Use X5R or X7R caps at minimum
VIN
EN/UV
SYNC
TR/SS
INTVCC
RT
PGND
GND
BST
SW
BIAS
PG
FB
NC
C5
100nF
L1
10uH
2x 1210 Tantalum Caps
C7
47uF
C8
47uF
+12V
TP5
TestPoint
R3
100k
R4
1M
C6
10pF
R5
88.7k
TP3
TestPoint
12V Power Good
GND
Switching Reg - Vbat to +5V
(high current switching reg.)
U4
LT8612
TP6
C12
4.7uF
C13
10nF
C14
1uF
R7
18.2k
C19
100nF
L3
2.5uH
C21
47uF
C22
47uF
+5V
TP10
R10
100k
R11
1M
C20
10pF
R12
243k
TP9
5V Power Good
Switching Reg - Vbat to 3.3V
(high current switching reg.)
U3
LT8612
satisfies voltage ripple condition of VDDIN for micrcontroller
by keeping voltage ripple to <10mV
has low ripple burst mode, but then limited to 700mA
TP4
active high
C9
4.7uF
C10
10nF
C11
1uF
R6
18.2k
Use X5R or X7R caps at minimum
C0G not bad
C15
100nF
L2
1.8uH
2x 1210 Tantalum Caps
C16
47uF
C18
47uF
TP8
+3V3
R8
1M
C17
4.7pF
R9
412k
TP7
3V3 Power Good
best high current low noise switching regulator with SEE test data
Board Power Monitor
I2C Address: 1000000
+3.3V
R1
0.002
Use high watt shunt!
U1
INA219AxDCN
IN+
IN-
VS
GND
SDA
SCL
A1
A0
I2C0_SDA
I2C0_SCL
C4
100nF
Debug LEDs
+12V
+5V
+3.3V
D1
RED
D2
RED
D3
RED
R13
330
R14
330
R15
330
Sheet: /Power Tree/
File: power_management.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 2/8

Decoupling Capacitors

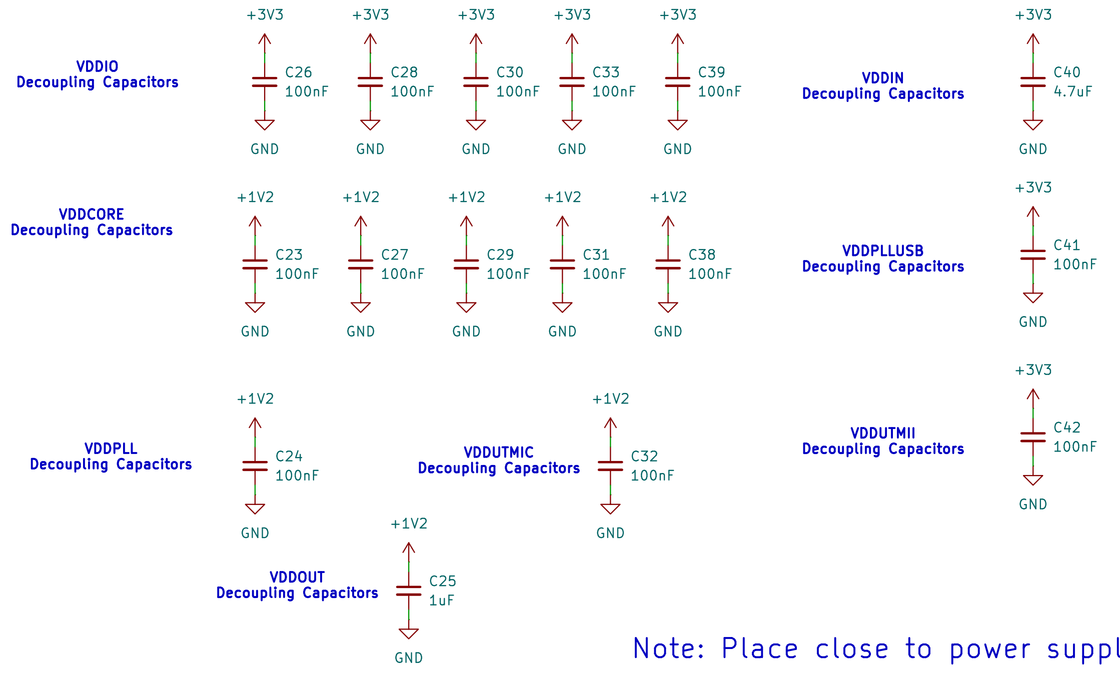


Table 7-1. Power Supplies

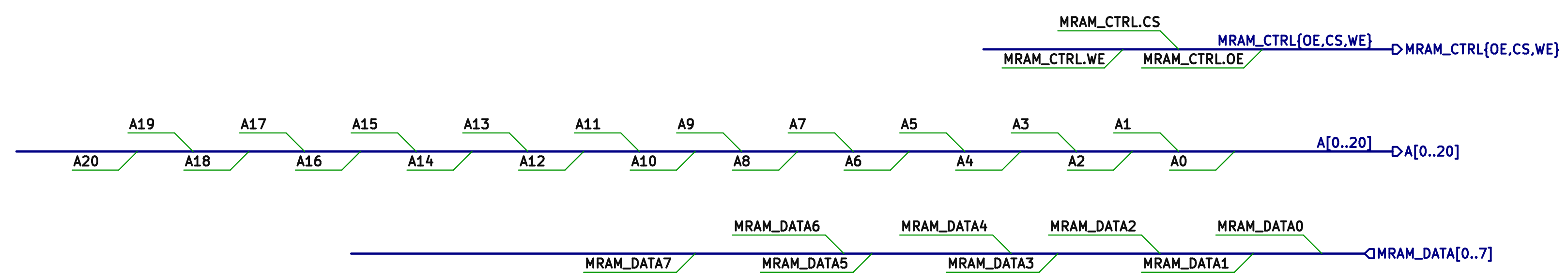
Name	Associated Ground	Powers
VDDCORE	GND	Core, embedded memories and peripherals.
VDDIO	GND	Peripheral I/O lines (Input/Output Buffers), backup part, 1 Kbytes of backup SRAM, 32 kHz crystal oscillator, oscillator pads. For USB operations, VDDIO voltage range must be between 3.0V and 3.6V.
VDDIN	GND, GNDANA	Voltage regulator input. Supplies also the ADC, DAC, and analog voltage comparator.
VDDPLL	GND, GNDPLL	PLLA and the fast RC oscillator.
VDDPLLUSB	GNDPLLUSB	UTMI PLL and 3 MHz to 20 MHz oscillator.
VDDUTMII	GNDUTMI	USB transceiver interface. Must be connected to VDDIO.
VDDUTMIC	GNDUTMI	USB transceiver core.

Table 58-5. Voltage Regulator Characteristics

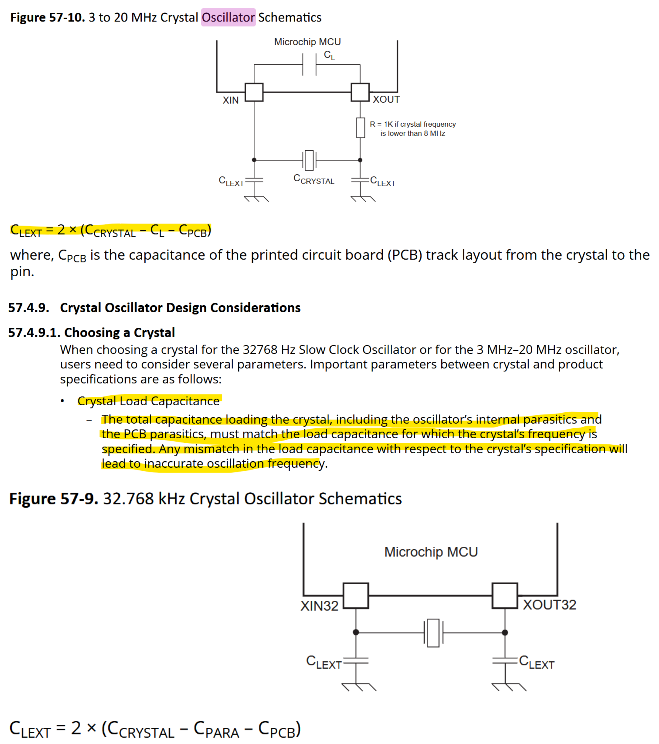
Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
V_{DDOUT}	DC Output Voltage	Normal mode, I_{LOAD} = 100 mA	1.2	1.23	1.26	V
		Standby mode	–	0	–	
I_{LOAD}	Maximum DC Output Current		–	–	150	mA
C_{DIN}	Input Decoupling Capacitor	(1)	–	4.7	–	µF
C_{DOUT}	Output Decoupling Capacitor	(2)	–	1	–	µF
		ESR	–	–	2	Ohm
t_{ON}	Turn-on Time	C_{DOUT} = 1 µF, V_{DDOUT} reaches DC output voltage	–	1	2.5	ms

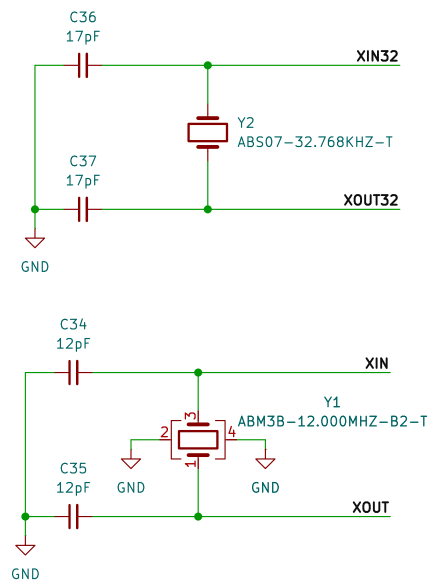
Note: Place close to power supply pins

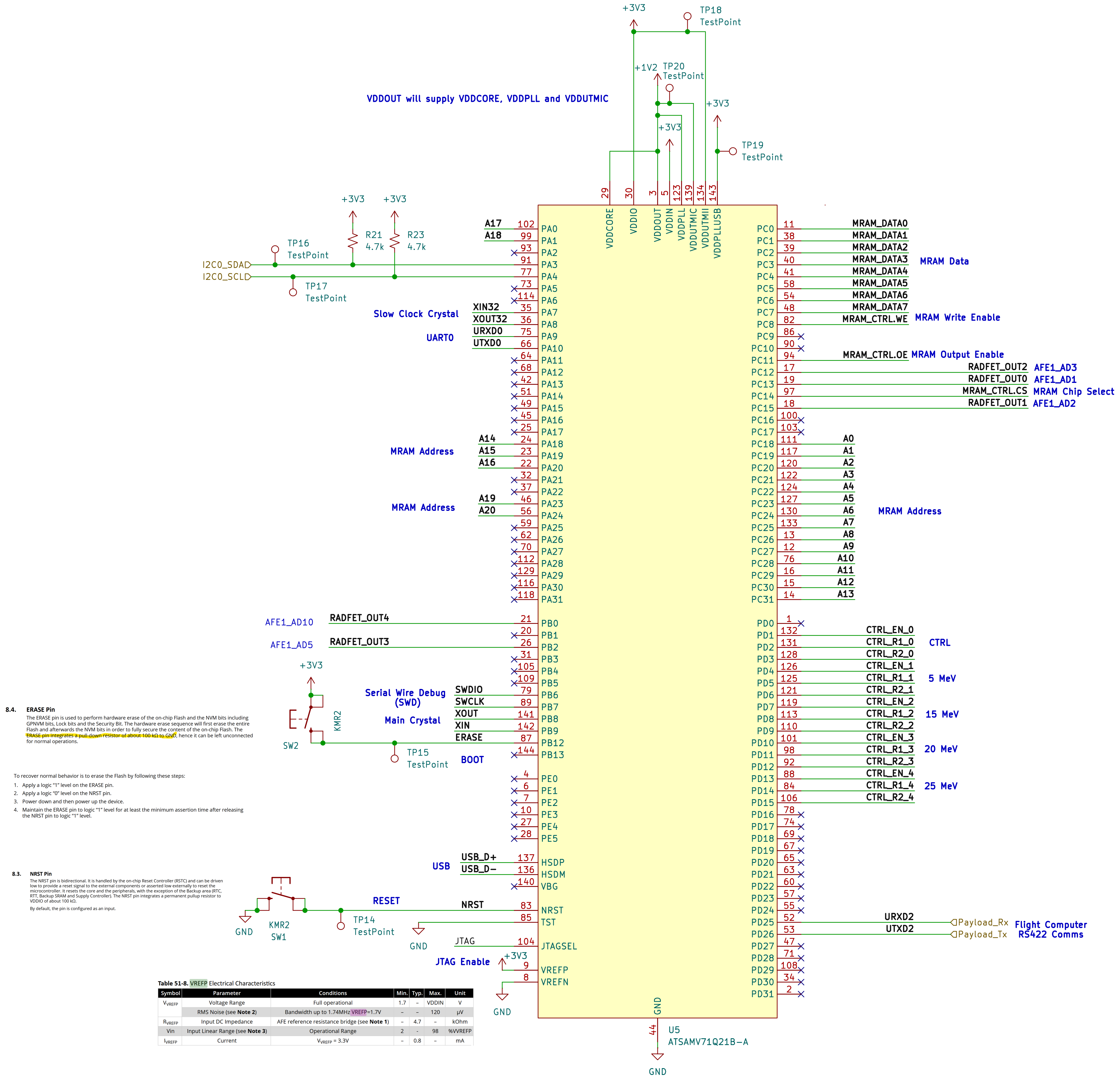
MRAM Bus Lines



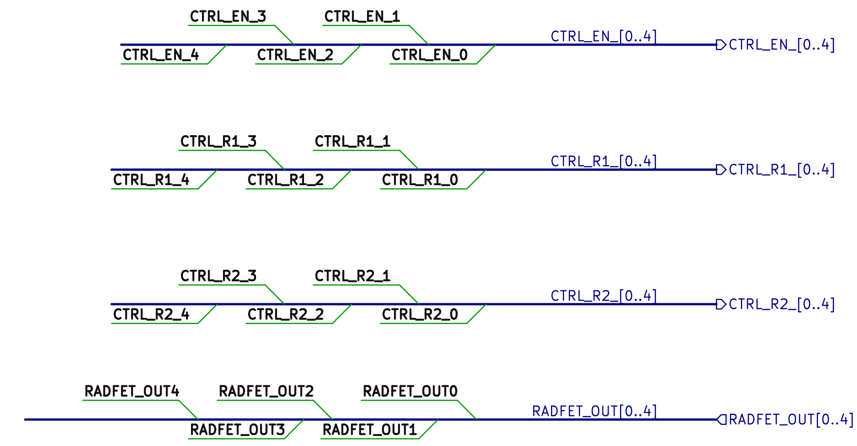
Crystal Oscilllator



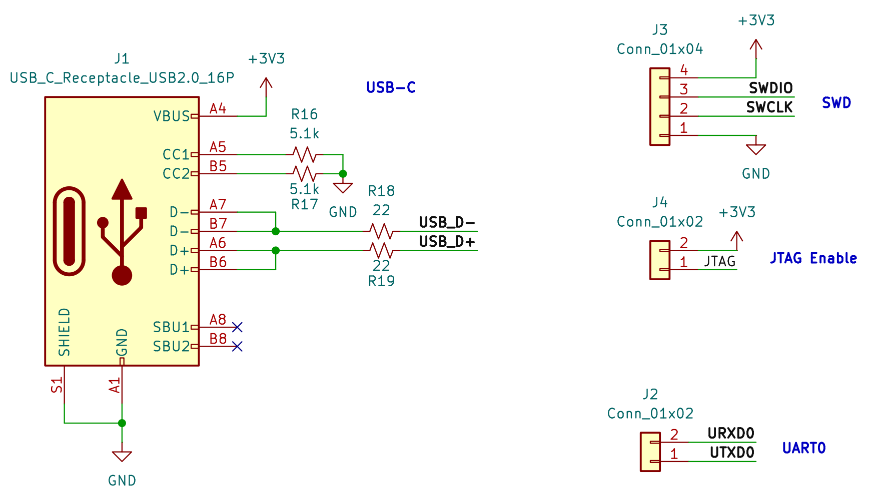




RadFET Bus Lines



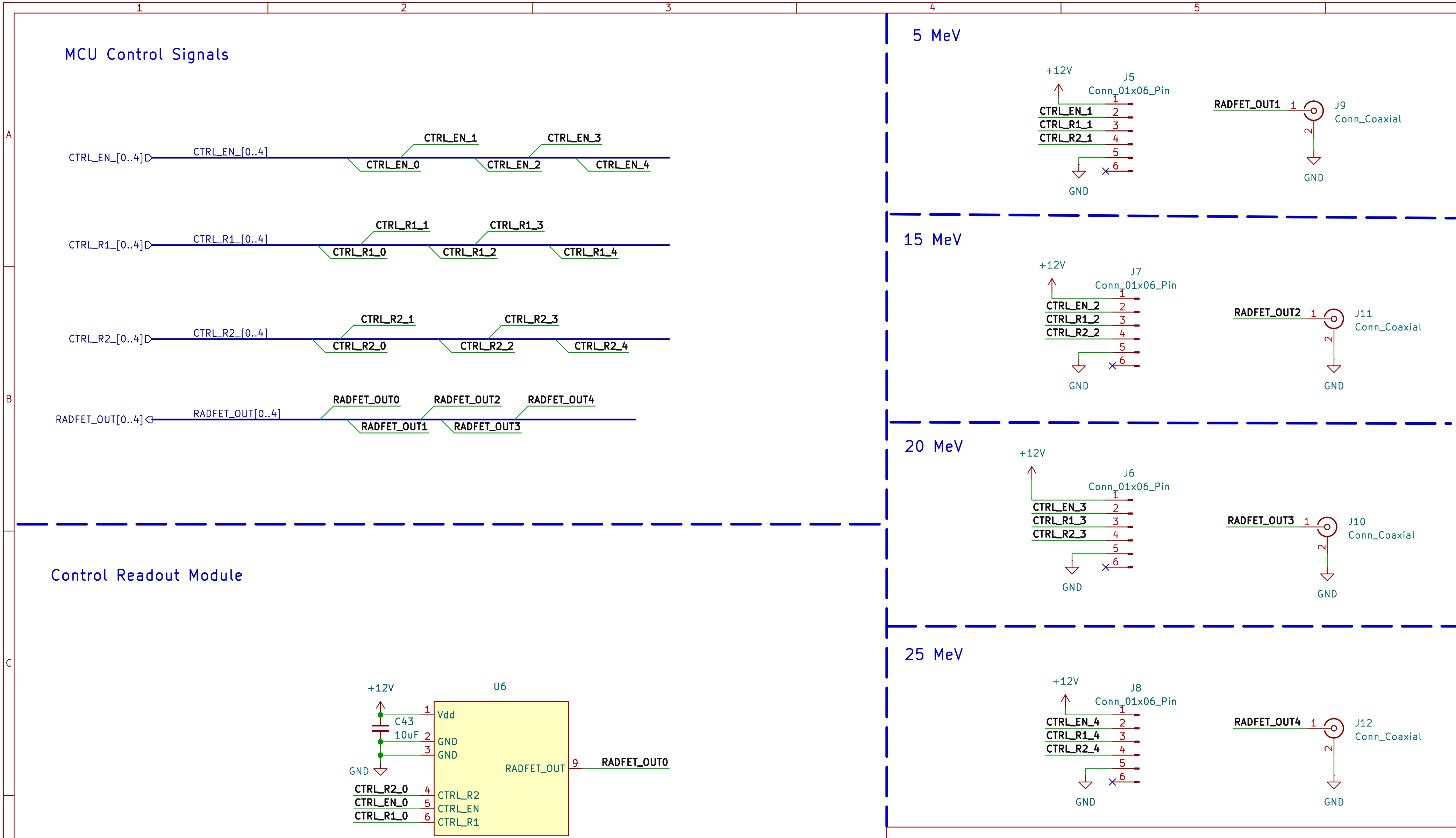
Debug Section



Sheet: /Microcontroller/
File: microcontroller.kicad_sch

Title:

Size: A2 Date: Rev:
KiCad E.D.A. 9.0.0 Id: 3/8



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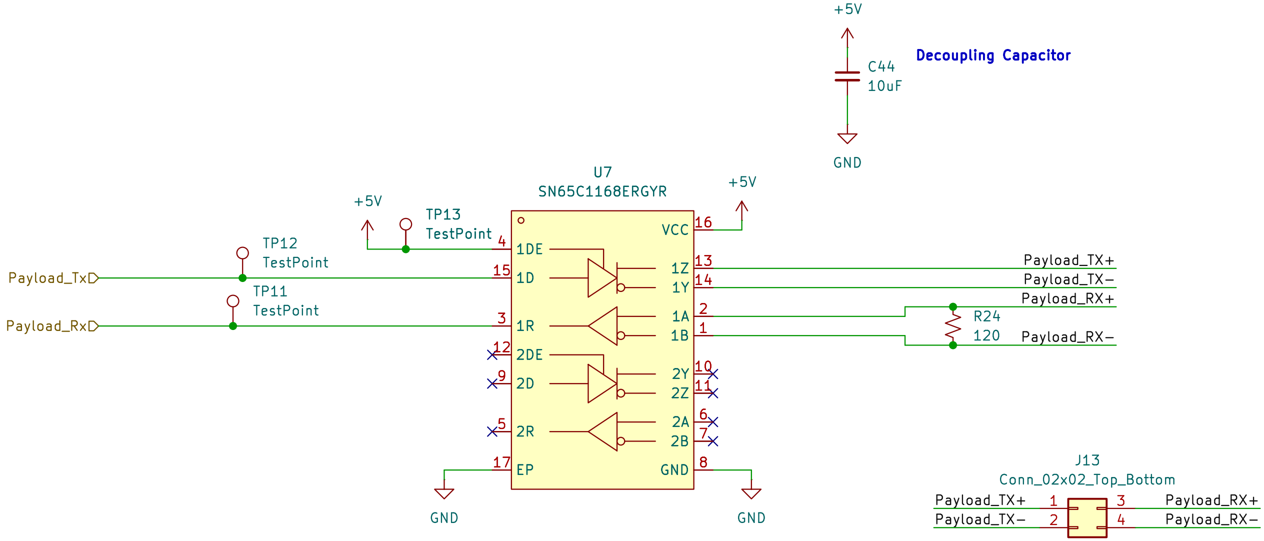
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 4/8



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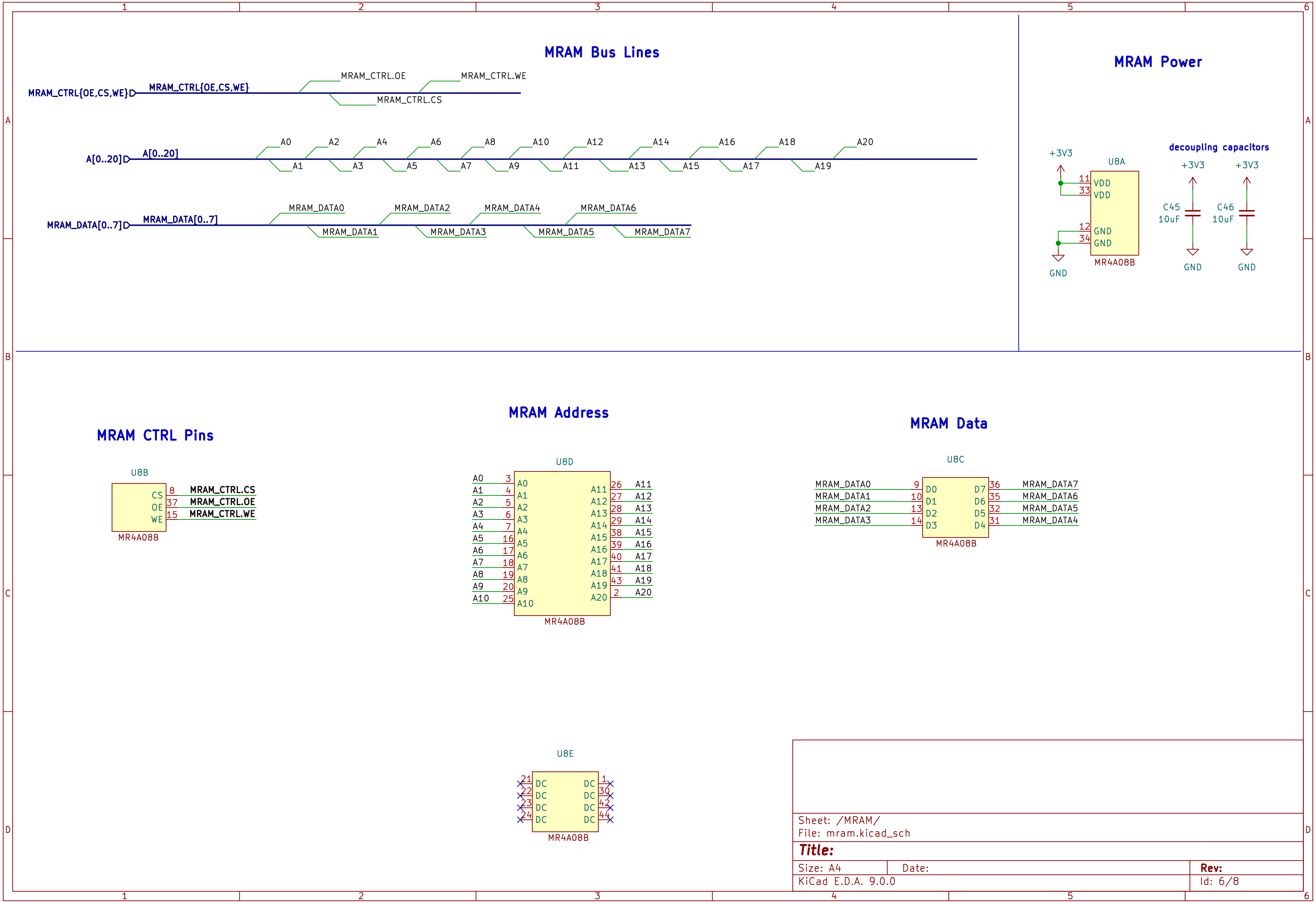
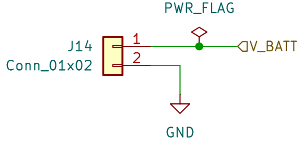
Size: A5	Date:	Rev:
KiCad E.D.A. 9.0.0		Id: 5/8


MRAM Bus Lines
MRAM_CTRL{OE,CS,WE}
MRAM_CTRL.OE
MRAM_CTRL.WE
MRAM_CTRL.CS
A[0..20]
A0 A2 A4 A6 A8 A10 A12 A14 A16 A18 A20
A1 A3 A5 A7 A9 A11 A13 A15 A17 A19
MRAM_DATA[0..7]
MRAM_DATA0 MRAM_DATA2 MRAM_DATA4 MRAM_DATA6
MRAM_DATA1 MRAM_DATA3 MRAM_DATA5 MRAM_DATA7
MRAM Power
+3V3
U8A
VDD
GND
MR4A08B
decoupling capacitors
C45 10uF
C46 10uF
GND
MRAM CTRL Pins
U8B
CS OE WE
MRAM_CTRL.CS
MRAM_CTRL.OE
MRAM_CTRL.WE
MRAM Address
U8D
MRAM Data
U8C
D0 D1 D2 D3 D7 D6 D5 D4
U8E
DC
Sheet: /MRAM/
File: mram.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 6/8

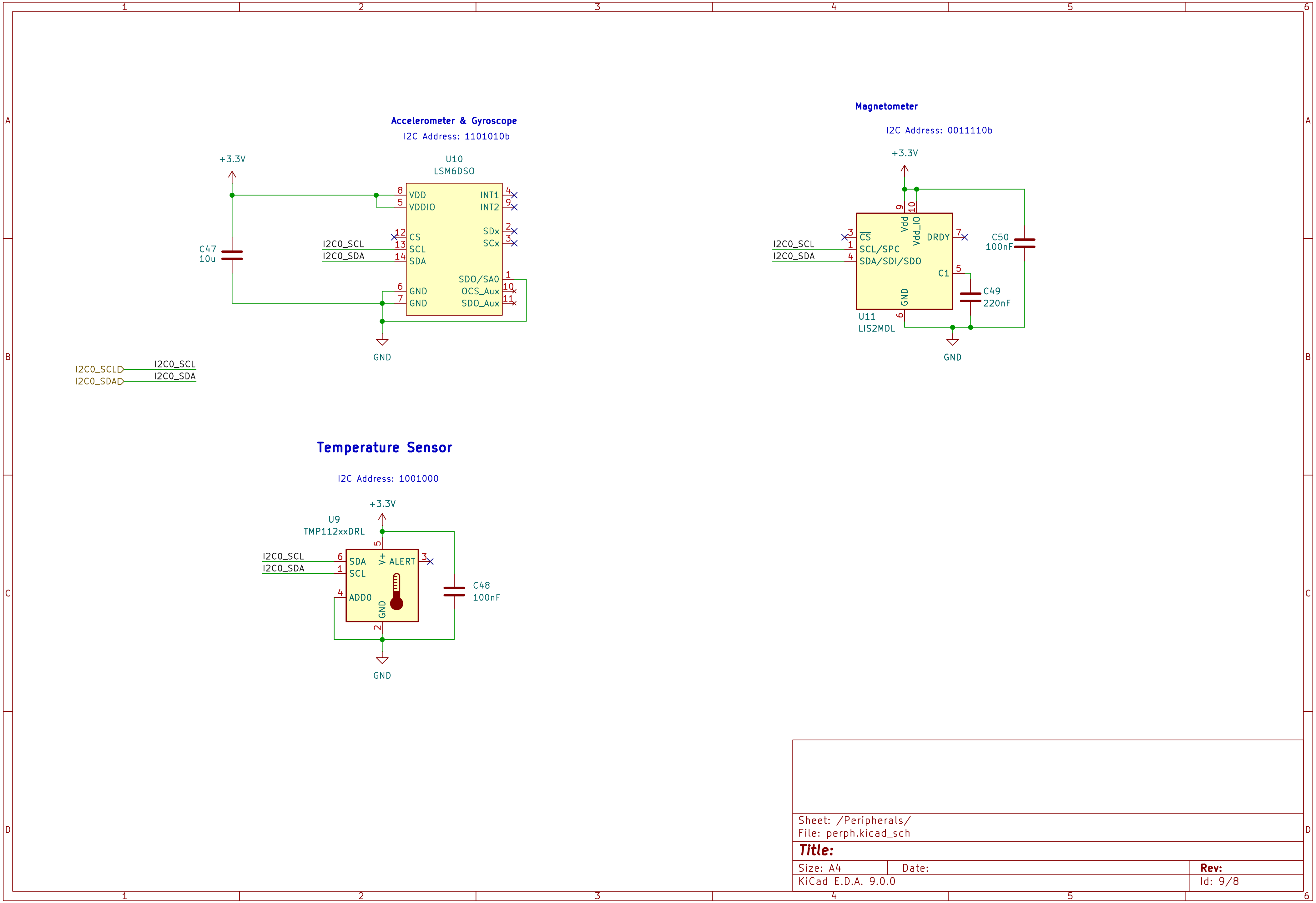


Sheet: /Power Input/
File: pwr_in.kicad_sch

Title:

Size: A5 | Date: | **Rev:**

KiCad E.D.A. 9.0.0 | Id: 7/8



Accelerometer & Gyroscope
I2C Address: 1101010b
U10
LSM6DSO
+3.3V
VDD
VDDIO
INT1
INT2
CS
SCL
SDA
SDx
SCx
SDO/SA0
GND
GND
OCS_Aux
SDO_Aux
I2C0_SCL
I2C0_SDA
C47
10u
GND
Magnetometer
I2C Address: 0011110b
+3.3V
Vdd
Vdd_IO
CS
SCL/SPC
SDA/SDI/SDO
DRDY
C1
GND
U11
LIS2MDL
C50
100nF
C49
220nF
GND
I2C0_SCL
I2C0_SDA
Temperature Sensor
I2C Address: 1001000
+3.3V
U9
TMP112xxDRL
SDA
SCL
ADD0
V+
ALERT
GND
C48
100nF
GND
Sheet: /Peripherals/
File: perph.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.0
Id: 9/8